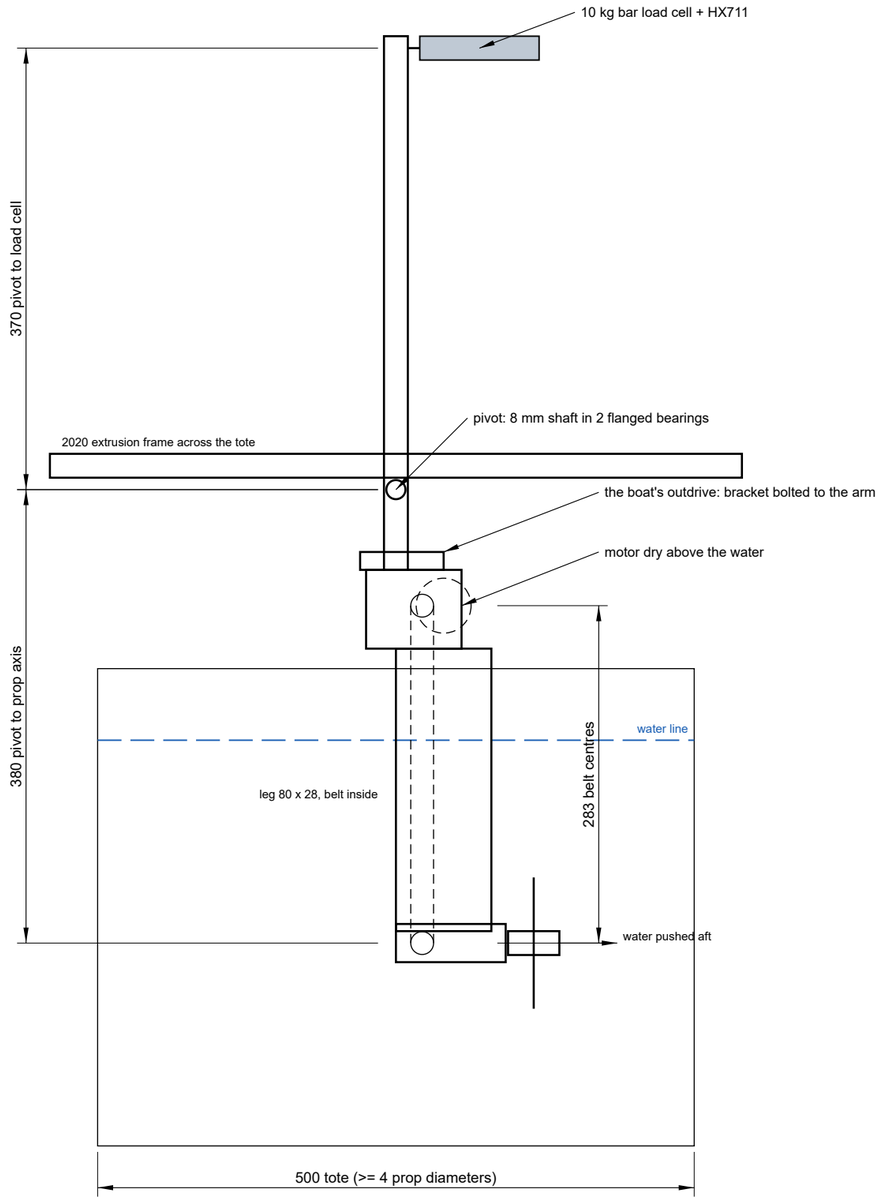


SIDE VIEW

SCALE 1:5

HOW IT WORKS

Thrust T at the prop axis turns the arm about the pivot; the arm's top presses the load cell. With equal arms the load-cell force equals T (F = T x 380 / 370 here).

PROCEDURE

1. Calibrate: hang known weights from a string over a pulley at the prop axis.
2. Power one outdrive from one M18 pack through a VESC; log current, voltage and rpm with VESC Tool's real-time data, and thrust from the HX711 (Pico or Arduino).
3. For each prop, step the rpm 1000 -> 4000 in 500 rpm steps, 10 s per step.
4. Plot thrust vs electrical power. Compare props at the same thrust:
~5 N (1 m/s cruise), ~10 N (1.5 m/s), ~25 N (3 m/s) per drive.
5. Weed test: hang loose strands upstream at 1500 rpm for 2 min; count wraps.

Keep the prop >= 1.5 D below the surface and >= 2 D from the walls and floor; a tote gives fair comparisons between props, not absolute numbers.
A plain 110 mm RC boat prop = baseline. Record water temperature and belt tension.

PARTS

2020 extrusion (~1.5 m) + corner brackets; 8 mm shaft + 2x KFL08 bearings; 10 kg bar load cell + HX711; 27-gal tote; one of the boat's outdrives, a VESC, one M18 pack, 20 A blade fuse and a kill switch.

DATA SHEET (one per prop)

rpm	V	A	W	thrust N	g/W	notes
1000						
1500						
2000						
...						
4000						

D.R.E.A.M.S V2 — autonomous survey catamaran

Prop thrust stand: pivot arm, load cell, test procedure

All dimensions mm unless noted. Third-angle projection. Do not scale the print; build from the numbers.

SCALE 1:5	SHEET 1 OF 1	REV B (belt outdrive)
DATE 2026-09-27	DRAWN Claude Code for M. Hong	SRC mechanical/toroidal_prop.py